

Week 2, part 1 of 1

Data

Lecture 2.1

1. Raw and tidy data
2. Data sanitisation
3. Training and holdout
4. Storage and versioning
5. Demo: cleaning and splitting
6. Finding datasets

Raw data

- **Messy csvs.**
- **A corpus or pile of text.** Scraped documents, chat transcripts, production logs. Typically unlabelled and scraped.

Raw data should be unchanged from source

- Preserve source snapshots under controlled access.
- Record transformations independently.
- Regenerate derived data from the source and transformation code.
- *Exception*: Track all copies when deleting data, and prevent backups from restoring deleted records.

Tidying raw data

1. Each variable forms a column.
2. Each observation forms a row.
3. Each type of observational unit forms a table.¹

1. Hadley Wickham, "Tidy Data", *Journal of Statistical Software* 59(10), 2014, §2.3, p. 4. <https://doi.org/10.18637/jss.v059.i10> ↩

Common dataset issues

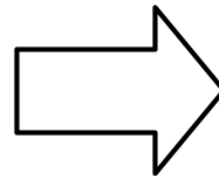
- Column headers are values, not variable names.
- Multiple variables are stored in one column.
- Variables are stored in both rows and columns.
- Multiple types of observational units are stored in the same table.
- A single observational unit is stored in multiple tables.¹

1. Wickham, "Tidy Data", §3, p. 5. The paper works each one through a real dataset and shows the fix. <https://doi.org/10.18637/jss.v059.i10> ↩

Tidying up an example dataset

```
tidy = wide.melt(id_vars="id",  
                 var_name="measurement", value_name="value")
```

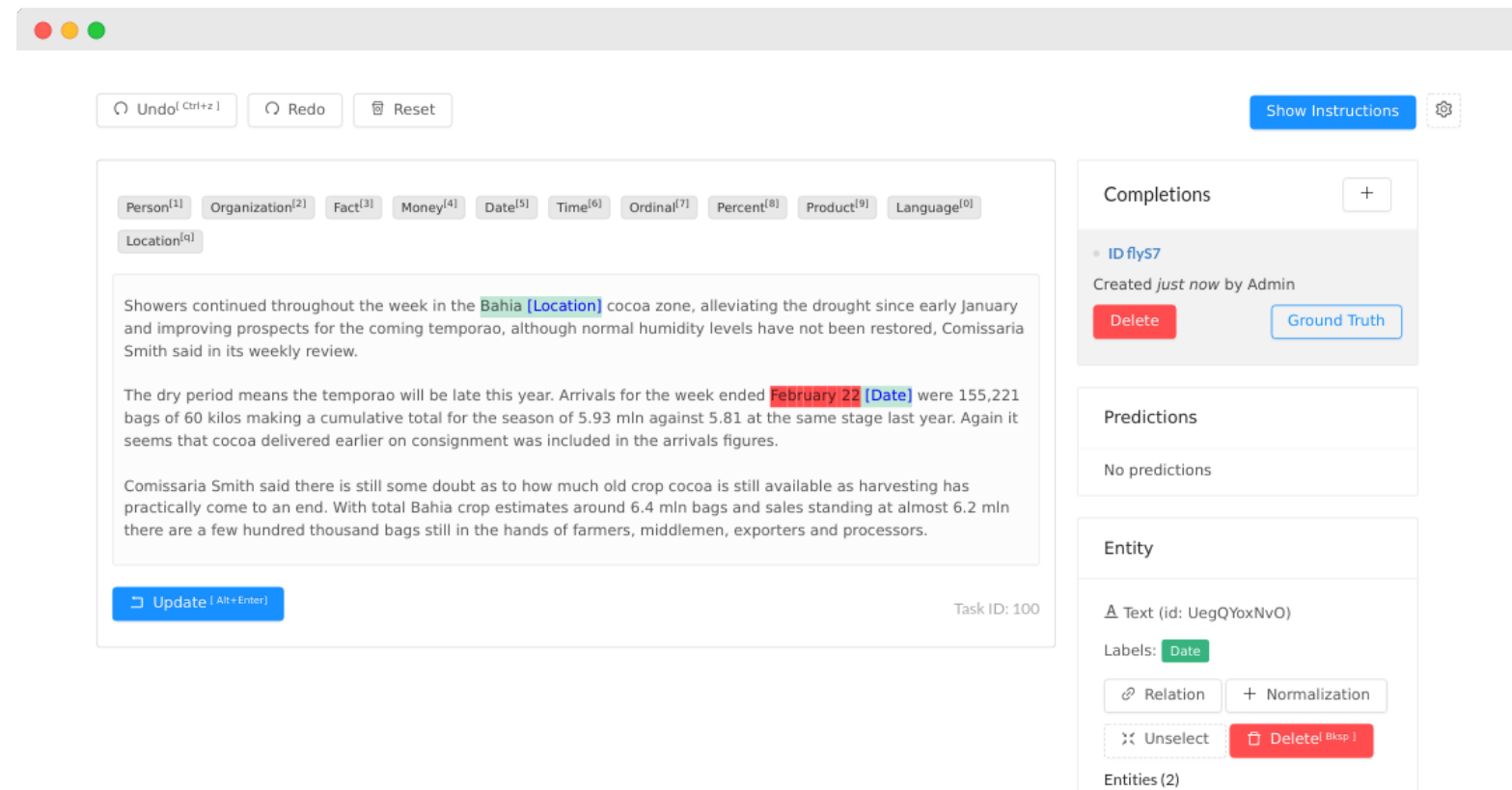
id	bp1	bp2
A	100	120
B	140	115
C	120	125



id	measurement	value
A	bp1	100
A	bp2	120
B	bp1	140
B	bp2	115
C	bp1	120
C	bp2	125

Look at your data

Read a hundred rows one at a time. Inspect and label them with something like Label Studio.



HumanSignal / Heartex, Label Studio. Apache-2.0

"You must remove all friction from the process of looking at data."¹

1. Hamel Husain, "Your AI Product Needs Evals", §"Looking At Your Traces". <https://hamel.dev/blog/posts/evals/> ↩

Sanitisation

Symptom	The job	Tool
Nulls where you expected values, a mark of 900	Structural validity	Pandera, Pydantic, dbt tests
Boilerplate, nav bars, keyword spam	Quality filtering	Your heuristics
The same document 40 times	Deduplication	MinHash, exact hash
Emails, API keys, someone's medical history	Things you may not keep	Presidio, scrubadub, deletion

Document the schema

"Feature expectations are captured in a schema."¹

```
import pandera.polars as pa

class Submission(pa.DataFrameModel):
    student_id: str =
pa.Field(str_matches=r"^z\d{7}$")
    mark: float = pa.Field(ge=0, le=100,
nullable=True)
    submitted_at: pa.DateTime = pa.Field(gt="2026-
09-14")
```

1. Breck, Cai, Nielsen, Salib and Sculley, "The ML Test Score", IEEE Big Data 2017, §II, Data 1. The other six data tests are in Table I. <https://research.google/pubs/the-ml-test-score-a-rubric-for-ml-production-readiness-and-technical-debt-reduction/> ↩

PII & Governance

- **PII.** Never "we will strip it later", always anonymise at ingest.
- **Secrets.** Scraped repositories and chat logs are full of live API keys.
- **Provenance.** Consider licence, source, date, terms of service.

ML Test Score Data 5: "The data pipeline has appropriate privacy controls."¹

1. Breck et al., "The ML Test Score", §II, Table I, Data 5. <https://research.google/pubs/the-ml-test-score-a-rubric-for-ml-production-readiness-and-technical-debt-reduction/> ↩

Holdouts (overfitting)

A model's score on data it was trained on tells you about its memory, not its ability.

Whatever you are building, you need a slice the model has never seen and you never tune against.

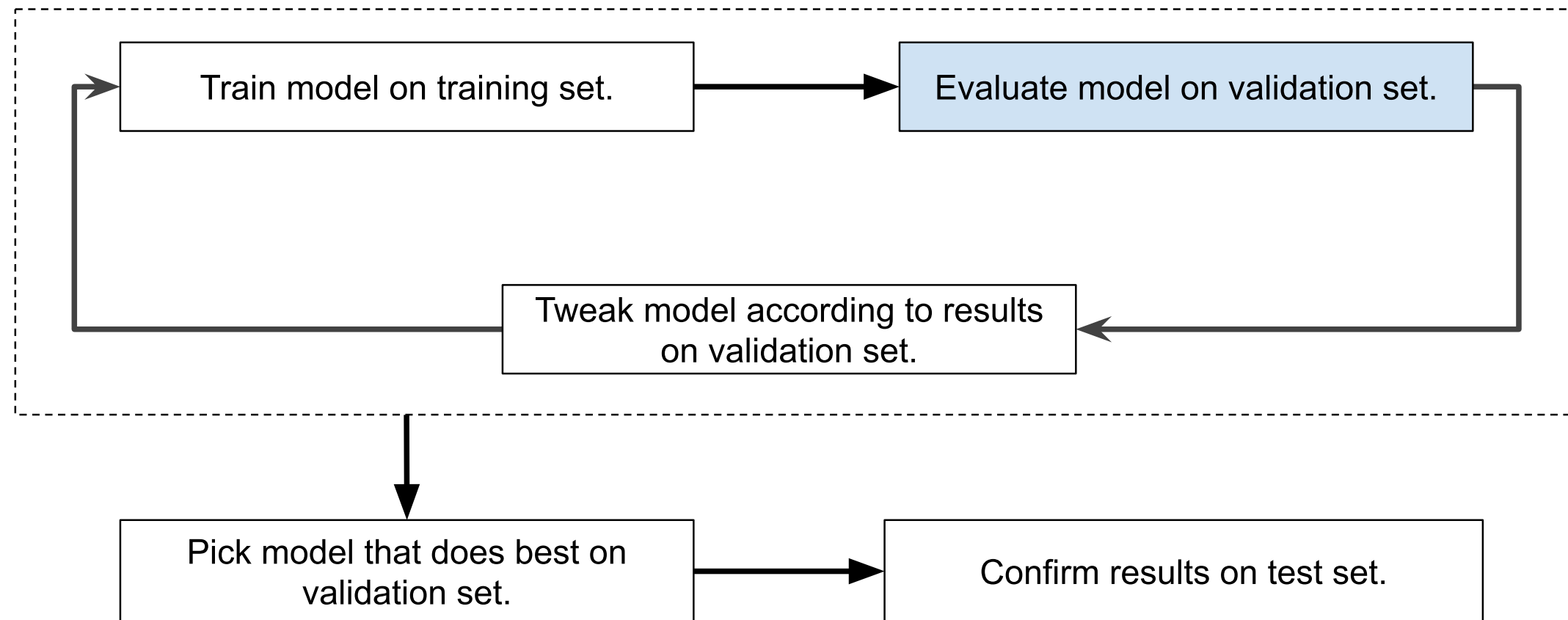
Be careful about randomly splitting your data!

- **Duplicates.** The same or near-identical examples appear in training and evaluation.
- **Groups.** All records for the same patient, customer or campaign stay together when evaluating generalisation to new entities.
- **Time.** Train on earlier data and evaluate on later data when predicting future events.¹

1. Google, *Rules of Machine Learning*, Rule #33, "Training-Serving Skew". <https://developers.google.com/machine-learning/guides/rules-of-ml>
↩

Training, development and final holdout

Validation = development. Test = final holdout.



Train and test how you serve

Rule #37: measure the skew.¹

Rule #29: save the exact set of
features used at serving time, and
feed those to training.

1. Google, *Rules of Machine Learning*, Rules #29 and #37. Compare applied-llms.org §2.1.1, "Check for development-prod skew": "The data we test our systems on during development should mirror what the systems will face in production."
<https://developers.google.com/machine-learning/guides/rules-of-ml> ↩

Storage: pick a format on purpose

Format	Good for	Bad at
CSV	Handing a human 200 rows	Types, nesting, size, quoting
JSONL	Append-only logs, ragged records	Scanning one field of many
Parquet	Anything you will query more than once	Appending a single row

Version data like code

- **Hugging Face Hub**: a git-backed private dataset repo. Commits, branches, revisions, etc.
- **DVC**: keeps a hash in git, the bytes in object storage, for data that lives outside the Hub.
- You can use UNSW OneDrive as a source for DVC repos!

Demo: cleaning and splitting

One spam dataset: inspect, validate,
deduplicate, compare row and group
splits.

`demos/2.1-data/demo.ipynb`

Finding datasets for your project

- **Government:** data.gov.au, Data.NSW, ABS statistics, Transport for NSW trips, Digital Earth Australia imagery.
- **Public:** Hugging Face, Research Data Australia, Atlas of Living Australia species records.
- **Join:** join sources by place/time, label documents, or archive permitted API snapshots.
- **Legality:** check original licences, API terms, privacy/consent and publication rights. Public access alone is insufficient.¹



1. Data.NSW: dataset-specific licences; Hugging Face dataset cards; OAIC: privacy and training generative AI. ↩

Building datasets

- **Collect your own:** opt-in prototype logs, authorised partner records, your own photos or phone sensors.
- **Find a few examples:** target missing cases. Use rules to propose labels or generate variants of verified training examples.
- **Synthetic datasets:** simulate task rules or generate questions from licensed documents. Check answers and mark synthetic rows.¹

1. dataset-generation $\hookleftarrow$

There are so many great tools for generating datasets, phyphox allows you to record sensor data and exports CSV.

